Milestone Progress Report: Lone Star Support

1. Executive Summary

The engineering team has successfully reached a key project milestone for **Lone Star Support**, establishing the core interactive mapping architecture, an AI-driven social media intelligence pipeline, telemetry and weather data integrations, and the foundational relational database schema. Direct client-side integrations currently power real-time visualization while background persistence services are being developed.

The project remains aligned with the technical roadmap. Tasks currently in progress—including background telemetry persistence, automated testing, and emergency infrastructure layers—are scheduled for full integration and delivery in the upcoming development cycle.

2. Completed During This Milestone

- **Interactive Geospatial Mapping**: Configured the MapLibre GL Texas map canvas featuring custom vector styling, interactive and clickable TxDOT incident markers, and mock spatial vectors for community incident reports.
- **AI Social Intelligence Pipeline**: Developed a Python 3.12+ ingestion worker connected to Bluesky Jetstream. The pipeline filters Texas-specific disaster keywords, executes structured triage and extraction via Gemini 3.5 Flash, resolves physical coordinates using Photon geocoding, and writes validated alerts to the Supabase public.social_alerts table.
- **Telemetry & Weather Services**: Integrated active alert querying from the National Weather Service (NWS), built 25-mile radius bounding box spatial queries for TxDOT road conditions, integrated ZIP code geocoding via Zippopotam.us, and delivered a

reusable WeatherSummaryCard front-end UI component.

- **Database Infrastructure**: Provisioned the Supabase instance and deployed the relational database schema optimized for geospatial query execution and rapid event ingestion.
- **UI/UX Design**: Created foundational system wireframes and visual workflows in Excalidraw, established the initial Next.js application shell, and configured layout frameworks using Tailwind CSS and custom component styling.

3. Current System Status

3.1 What Is Currently Working

- **Interactive Texas Map Canvas**: Real-time rendering of active road hazards and weather warnings directly from live TxDOT and NWS API feeds.
- **Bluesky AI Ingestion Engine**: Fully operational processing pipeline: Bluesky Jetstream → Keyword Filtering → Gemini 3.5 Flash Triage → Photon Geocoding → Supabase public.social_alerts storage.
- **Core Database Schema**: Fully modeled, configured, and running on Supabase with spatial indexing enabled.

3.2 What Is Not Yet Working

- **TxDOT & NWS Database Ingestion**: Automated scheduled background workers to transition data handling from client-side direct queries to persistent Supabase storage.
- **Emergency Infrastructure Layer**: Querying and rendering critical facilities (hospitals, shelters, fire stations) via the OpenStreetMap Overpass API.
- **Automated Security Testing**: Complete Playwright and Pytest end-to-end integration test coverage and REST API security verification.
- **Community Report Moderation**: Administrative moderation interfaces and automated content filter logic for user-submitted field reports.
- **Mobile Viewport Optimization**: Touch-optimized gesture controls, responsive

navigation drawers, and mobile screen layouts.

4. Major Challenges Encountered

- **NWS Alert Geometry Inconsistencies**: Certain National Weather Service advisories lack usable geospatial polygon coordinates, providing only text summaries or FIPS county lists. To display these non-geometries reliably without breaking MapLibre GL layers, the team is evaluating secondary county-boundary shape lookup tables.
- **MapLibre GL Integration Hurdles**: Initial development required resolving reactivity issues between React state lifecycle triggers, dynamic vector tile layers, and dynamic event listener attachment for hover card popups.
- **Automated Testing Setup Complexity**: Establishing full-stack test pipelines posed configuration hurdles across Next.js dynamic routes and Python background tasks. Coordinating Playwright for browser interactions and Pytest for API endpoints requires isolated environment orchestration to ensure complete security verification.

5. Changes Made to the Original Plan

- **Phased API Ingestion Approach**: To validate UI responsiveness and geospatial rendering early, direct API fetching for TxDOT and NWS feeds was implemented on the frontend. Transitioning these tasks to background database workers was shifted to the immediate next sprint without blocking map development.
- **Scope & Architectural Fidelity**: Core project scope, component definitions, and system architecture remain unchanged and fully aligned with the initial project proposal.

6. Next Steps & Upcoming Deliverables

1. **TxDOT & NWS Ingestion Engine**: Finalize automated Python background workers to persist, normalize, and index state telemetry in Supabase.

2. **Overpass API Integration**: Implement spatial querying to render localized emergency infrastructure (hospitals, fire stations, temporary shelters).
3. **Automated Security & Regression Testing**: Complete setup of Playwright end-to-end tests and Pytest suites for API authentication and endpoint validation.
4. **Community Report Moderation**: Build moderation pipelines and rules for automated content filtering of user-submitted reports.
5. **Mobile Viewport Responsiveness**: Apply mobile-first styling, touch controls, and compact UI views for handheld devices.

7. Individual Contributions

Subsystem	Feature Description	Status	Primary Owner
Telemetry	NWS Active Alert Querying	Completed	Kevin Hernandez
Telemetry	TxDOT 25-Mile Radius Bounding Query	Completed	Kevin Hernandez
Telemetry	ZIP Code Geocoding via Zippopotam.us	Completed	Kevin Hernandez
Telemetry	Reusable	Completed	Kevin Hernandez

Subsystem	Feature Description	Status	Primary Owner
	WeatherSummaryCard Component		
Testing	Automated Testing Framework (Playwright/Pytest)	In Progress	Kevin Hernandez
AI Pipeline	Bluesky Jetstream Consumer and Keyword Filter	Completed	Emilio Villar
AI Pipeline	Gemini 3.5 Flash Summarization & Triage	Completed	Emilio Villar
AI Pipeline	Photon Geocoding Integration	Completed	Emilio Villar
Database	Supabase Database Configuration	Completed	Emilio Villar
Database	Database Ingestion	In Progress	Emilio Villar

Subsystem	Feature Description	Status	Primary Owner
	for TxDOT and NWS		
Map	MapLibre GL Map Canvas and Layer Setup	Completed	Julian Spindola
Map	Clickable TxDOT Incident Rendering	Completed	Julian Spindola
Map	Community Report Mock Markers	Completed	Julian Spindola
UI / UX	Foundational Wireframes (Excalidraw)	Completed	Mariana Lozano
UI / UX	Initial Application Layout and Component Styling	Completed	Isabel Villareal
Mobile	Viewport Responsiveness	In Progress	Isabel Villareal

Subsystem	Feature Description	Status	Primary Owner
	and Touch Controls		

Summary of Individual Contributions

- **Kevin Hernandez (Telemetry & Testing Lead)**: Engineered telemetry fetch modules for NWS active alerts, TxDOT 25-mile bounding queries, ZIP geocoding services, and the WeatherSummaryCard UI component; currently leading automated testing setup across Playwright and Pytest framework layers.
- **Emilio Villar (AI Pipeline & Database Lead)**: Built the Bluesky Jetstream ingestion client, Gemini 3.5 Flash triage and summary pipeline, Photon geocoder integration, and core Supabase schema; currently developing persistent background workers for TxDOT and NWS feeds.
- **Julian Spindola (Mapping Lead)**: Architected the MapLibre GL canvas, layer hierarchy, clickable TxDOT incident interactions, and initial mock markers for community field reports.
- **Mariana Lozano (UI/UX Design)**: Created comprehensive Excalidraw wireframes establishing system user flows, component positioning, and dashboard layout structure.
- **Isabel Villareal (Frontend & Mobile Lead)**: Developed the primary Next.js dashboard frame, visual styling, and component hierarchy; currently optimizing viewport layouts and gesture controls for mobile devices.